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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER | | | | | |

CANDIDATE NUMBER | | | | |

9250109413

ADDITIONAL MATHEMATICS **4037/12**

Paper 1 **May/June 2011**

2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
The use of an electronic calculator is expected, where appropriate.
You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.
The total number of marks for this paper is 80.

For Examiner's Use	
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Total	

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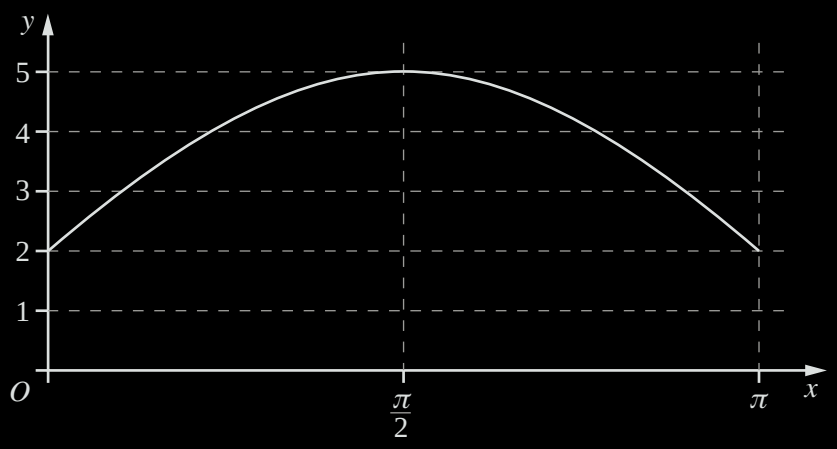
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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

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The figure shows the graph of $y = k + m\sin px$ for $0 \leqslant x \leqslant \pi$, where k, m and p are positive constants. Complete the following statements.

k = ... m = ... p = ... [3]

(b) The function g is such that $g(x) = 1 + 5\cos 3x$. Write down

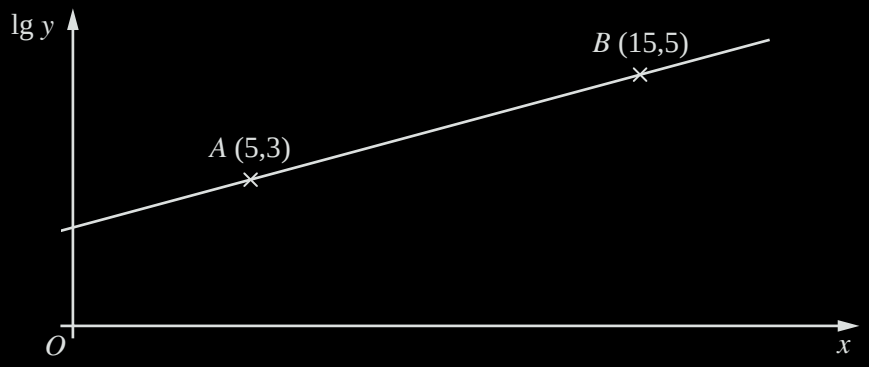
(i) the amplitude of g, [1]

(ii) the period of g in terms of π. [1]

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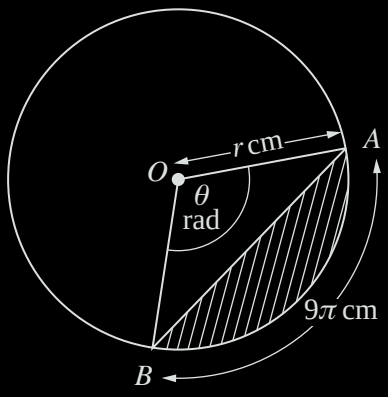
(i) Express $\lg y$ in terms of x. [3]

(ii) Show that $y = a(10^{bx})$ where a and b are to be found. [3]

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(i) Show that $\theta = \frac{3\pi}{5}$. [2]

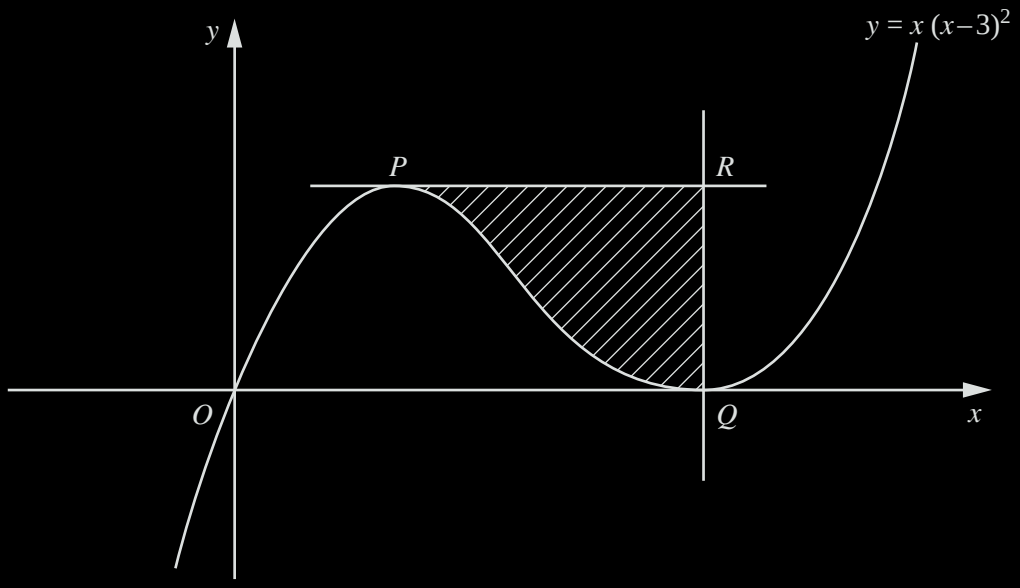
(ii) Find the value of r. [2]

(iii) Find the area of the shaded region. [3]

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The diagram shows the curve $y = x(x - 3)^2$. The curve has a maximum at the point P and touches the x-axis at the point Q. The tangent at P and the normal at Q meet at the point R. Find the area of the shaded region PQR. [10]



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